HANDS-ON LAB

Navigate the AWS ecosystem, match services to real-world problems, and reason about *why* each service exists — not just *what* it does.

~90 minutes Beginner Console + CLI (read-only)

YOUR NAME e.g. Ada Lovelace

0 Overview & Learning Outcomes

In this lab you **will not build anything** — you will *explore, compare, and reason*. The goal is to build a mental map of AWS services so that when you face a real architecture decision, you already know where to look.

By the end you can

1. Navigate the AWS Console service catalog and find key information about any service in under 60 seconds.
2. Use the AWS CLI to discover regions, services, and basic resource metadata.
3. Explain **why** a service exists — what problem it solves — not just its feature list.
4. Choose appropriate AWS services for a given business scenario.
5. Distinguish services within the same family (e.g. S3 vs EBS vs EFS).

Prerequisites

- AWS Console access (IAM user or SSO) — **read-only is fine**.
- AWS CLI installed and configured (`aws configure list` shows credentials).
- Completed the lesson on *Cloud Computing Models* and *AWS Services*.

1 Activity 1 — Console Safari 20 min

Open the **AWS Management Console**. Your mission: speed-navigate through six services and collect intel.

Instructions

1. Click the "**Services**" dropdown (top-left hamburger menu). Observe how AWS groups services by category.
2. For **each service below**, navigate to its Console page, spend ~90 seconds reading the landing dashboard, then type your answer. A few questions need one extra click beyond the landing page — the exact click is spelled out in the question.

EC2 — CLICK "LAUNCH AN INSTANCE" (YOU DON'T HAVE TO FINISH OR SUBMIT IT). UNDER "INSTANCE TYPE", NAME TWO INSTANCE TYPE FAMILIES YOU SEE IN THE DROPDOWN (E.G. T2, T3, M5…).

Type your answer...

S3 — OPEN ANY BUCKET (OR CREATE ONE) → CLICK "UPLOAD" → "ADD FILES" → EXPAND THE "PROPERTIES" SECTION → OPEN THE "STORAGE CLASS" DROPDOWN. HOW MANY STORAGE CLASSES ARE LISTED? NAME AT LEAST 3. (YOU CAN CANCEL THE UPLOAD AFTERWARD — NO NEED TO FINISH IT.)

Type your answer...

RDS — WHICH DATABASE ENGINES ARE AVAILABLE WHEN YOU CLICK "CREATE DATABASE"?

Type your answer...

LAMBDA — WHAT RUNTIMES (LANGUAGES) CAN YOU CHOOSE WHEN CREATING A FUNCTION?

Type your answer...

VPC — HOW MANY VPCS ALREADY EXIST IN YOUR ACCOUNT IN THE CURRENT REGION? WHAT'S THE CIDR?

Type your answer...

CLOUDWATCH — CLICK "DASHBOARDS" IN THE LEFT MENU, THEN "ALARMS" → "ALL ALARMS." WHAT DO YOU SEE IN EACH? (NOTE THE EMPTY STATE IF NONE.)

Type your answer...

> **Why this matters:** In production, you'll be dropped into unfamiliar AWS accounts constantly. Knowing how to orient yourself in the Console in seconds — not minutes — is a real skill.

2 Activity 2 — CLI Discovery

15 min ▼

The Console is visual, but the CLI reveals things the Console hides. Run these commands and answer the questions.

Step 1 — Regions & Availability Zones

```
# List all AWS regions
aws ec2 describe-regions --output table

# List AZs in your current region
aws ec2 describe-availability-zones --output table
```

HOW MANY REGIONS EXIST? HOW MANY AZS DOES YOUR CURRENT REGION HAVE? DO ALL REGIONS HAVE THE SAME NUMBER?

Type your answer...

Step 2 — What's running?

```
# Any EC2 instances running?
aws ec2 describe-instances \
  --query "Reservations[].Instances[].{ID:InstanceId,State:State.Name,Type:InstanceType}"
  --output table

# Any S3 buckets?
aws s3 ls

# Any Lambda functions?
aws lambda list-functions --output table
```

DID YOU FIND ANYTHING ALREADY DEPLOYED? IF SO, WHO OR WHAT CREATED IT?

Type your answer...

Step 3 — Service help

```
# Discover all CLI sub-commands for a service
aws s3api help | head -40
aws dynamodb help | head -40
```

PICK ONE SUB-COMMAND YOU DIDN'T KNOW EXISTED. WHAT DO YOU THINK IT DOES?

e.g. s3api list-bucket-metrics-configurations — I think it…

> **Think about it:** Why would you use `aws s3api` vs `aws s3`? The `s3` commands are high-level shortcuts; `s3api` maps 1:1 to the S3 REST API — useful when you need fine-grained control or automation.

3 Activity 3 — "Why Does This Service Exist?"

35 min ▼

This is the core of the lab. You'll read **four business scenarios** and decide which AWS services solve each problem — and more importantly, *why*.

> **Rules:** For each scenario, (a) pick the AWS services you'd use, (b) write one sentence explaining *what problem that service solves* in this context, and (c) name one alternative you considered and why you rejected it.

SCENARIO A — THE STARTUP BLOG

A two-person startup needs to host a blog. They get ~500 visitors/day. They want it cheap, and they don't want to manage servers. The blog is static HTML/CSS/JS generated by a build tool.
Hint: Think storage + CDN. Do they need EC2?

YOUR ANSWER — SCENARIO A

Services chosen:

e.g. S3, CloudFront

Why these services?

What problem does each solve here?

Rejected alternative & reason:

e.g. EC2 — because…

SCENARIO B — THE E-COMMERCE PEAK

An online store has steady traffic most of the year but gets 10× traffic during Black Friday. Their MySQL database is the bottleneck. They can't afford downtime, and the database has to stay relational.

Hint: Think managed database + read replicas + caching. What's the difference between RDS, Aurora, and ElastiCache here?

YOUR ANSWER — SCENARIO B

Services chosen:

e.g. Aurora, ElastiCache

Why these services?

What problem does each solve here?

Rejected alternative & reason:

e.g. DynamoDB — because...

SCENARIO C — THE IMAGE PROCESSING PIPELINE

A photography company uploads ~2,000 images per day to a storage bucket. Each image needs to be resized into 3 thumbnails automatically. They don't want to run servers 24/7 waiting for uploads.

Hint: Think event-driven. What triggers what?

YOUR ANSWER — SCENARIO C

Services chosen:

e.g. S3, Lambda

Why these services?

What problem does each solve here?

Rejected alternative & reason:

e.g. EC2 with cron — because...

SCENARIO D — THE COMPLIANCE HEADACHE

A healthcare company in Germany must store patient records. EU regulations (GDPR) require data to stay in the EU. They need encryption at rest and in transit, audit trails for every access, and strict access controls. They want a managed database so they don't patch anything.

Hint: Think region choice + encryption + managed DB + logging. How many services do you need?

YOUR ANSWER — SCENARIO D

Services chosen:

e.g. RDS, KMS, CloudTrail, IAM

Why these services?

What problem does each solve here?

Rejected alternative & reason:

e.g. Self-managed MySQL on EC2 — because...

4 Activity 4 — Confusion Pairs

15 min

AWS has services that *sound* similar but solve different problems. For each pair, write one sentence that captures the **key difference**.

> **Quick primer** — you haven't met two of these services yet, so here's a one-line description of each to reason from:
>
> - **CloudFront:** AWS's CDN — caches copies of content at edge locations close to users worldwide.
> - **Global Accelerator:** routes traffic over AWS's private global network backbone using static anycast IPs, instead of the public internet.
> - **Secrets Manager:** stores sensitive values (like DB passwords) and can automatically rotate them.
> - **Systems Manager Parameter Store:** stores configuration values and secrets, but doesn't rotate them automatically.

S3 vs EBS

The key difference is...

RDS vs DynamoDB

The key difference is...

`EC2 vs Lambda`

The key difference is...

`CloudFront vs Global Accelerator`

The key difference is...

`Secrets Manager vs Parameter Store`

The key difference is...

> **Thinking prompt:** For each pair, ask yourself: "If I had to explain this to a non-technical PM in 10 words, what would I say?"

★ **Bonus Challenge:** Pick any two services NOT listed above that you find confusing.

e.g. ECS vs EKS

The key difference is...

5 Wrap-Up & Reflection

5 min ▼

WHICH SERVICE SURPRISED YOU THE MOST — AND WHY?

Type your answer...

WHAT'S ONE SERVICE YOU WANT TO USE HANDS-ON NEXT?

Type your answer...

Key Takeaways

- AWS has 200+ services, but ~90% of real-world workloads use fewer than 20. Learn the core trios first: **EC2 / ECS / Lambda** (compute), **S3 / EBS / EFS** (storage), **RDS / DynamoDB / ElastiCache** (databases).
- Always start from the **problem**, not the service. "I need to store objects cheaply" → S3. "I need sub-millisecond key-value lookups" → DynamoDB or ElastiCache.
- The CLI is your power tool for discovery and automation — get comfortable reading `help` output.

Next up: In the following labs you'll go hands-on with **IAM** — the service that controls who can do what across everything you explored today.

Save my answers

Export as Markdown